

Complex Numbers

ANSWER KEY



Basic arithmetic with complex numbers

- 1 Given $z_1 = 4 + 3i$ and $z_2 = 2 - 5i$, find $z_1 + z_2$ and $z_1 - z_2$.
 $z_1 + z_2 = (4 + 2) + (3 - 5)i = 6 - 2i$. $z_1 - z_2 = (4 - 2) + (3 - (-5))i = 2 + 8i$.
- 2 Find $z_1 z_2$ for $z_1 = 2 + 3i$ and $z_2 = 4 - i$.
 $z_1 z_2 = (2 + 3i)(4 - i) = 8 - 2i + 12i - 3i^2 = 8 + 10i + 3 = 11 + 10i$.

The complex conjugate and division

- 3 Find the conjugate of $z = 7 - 2i$, and compute $z\bar{z}$.
 $\bar{z} = 7 + 2i$. $z\bar{z} = (7 - 2i)(7 + 2i) = 49 - 4i^2 = 49 + 4 = 53$.
- 4 Simplify $\frac{3+i}{2-3i}$ by multiplying by the conjugate of the denominator.
 $\frac{3+i}{2-3i} \cdot \frac{2+3i}{2+3i} = \frac{(3+i)(2+3i)}{4+9} = \frac{6+9i+2i+3i^2}{13} = \frac{3+11i}{13} = \frac{3}{13} + \frac{11}{13}i$.

Modulus and the Argand diagram

- 5 Find the modulus $|z|$ of $z = 5 - 12i$.
 $|z| = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.
- 6 The graph shows several complex numbers plotted on the Argand diagram (real part horizontal, imaginary part vertical).
- a Read off the complex number plotted at $(4, 1)$ in Cartesian form. The point $(4, 1)$ corresponds to $4 + i$.
- b Find the modulus of the complex number plotted at $(-2, -3)$. The point $(-2, -3)$ corresponds to $-2 - 3i$, with modulus $\sqrt{(-2)^2 + (-3)^2} = \sqrt{13}$.

Solving equations with complex roots

- 7 Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.
By the quadratic formula: $x = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$.
- 8 One root of a quadratic equation with real coefficients is $3 - 4i$. State the other root, and find the quadratic equation.
Since the coefficients are real, complex roots occur in conjugate pairs, so the other root is $3 + 4i$. Sum of roots = 6; product = $(3 - 4i)(3 + 4i) = 9 + 16 = 25$. Equation: $x^2 - 6x + 25 = 0$.

- 9 Given that $z = 1 + 2i$ is a root of $z^3 - 4z^2 + 9z - 10 = 0$ (with real coefficients), find the other two roots.
 Since the coefficients are real, $z = 1 - 2i$ is also a root. These two roots multiply to $(1 + 2i)(1 - 2i) = 1 + 4 = 5$ and sum to 2, giving the quadratic factor $z^2 - 2z + 5$. Dividing the cubic by this factor: $z^3 - 4z^2 + 9z - 10 \div (z^2 - 2z + 5) = z - 2$, so the third root is $z = 2$. The other two roots are $1 - 2i$ and 2.

Converting to polar (modulus-argument) form

- 10 Write $z = \sqrt{3} + i$ in polar form $r(\cos \theta + i \sin \theta)$, giving the argument in radians.
 $r = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$. $\theta = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$. Polar form: $z = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$.
- 11 Convert $z = 6\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$ to Cartesian form.
 $a = 6\cos\left(\frac{2\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$. $b = 6\sin\left(\frac{2\pi}{3}\right) = 6\left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$. So $z = -3 + 3\sqrt{3}i$.

Multiplying and dividing in polar form

- 12 Given $z_1 = 4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ and $z_2 = 2\left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12}\right)$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.
 Multiply moduli, add arguments: $z_1 z_2 = 8\left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12}\right)$. Divide moduli, subtract arguments: $\frac{z_1}{z_2} = 2\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$.
- 13 Given $z_1 = 5\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$ and $z_2 = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.
 $z_1 z_2 = 5\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$. $\frac{z_1}{z_2} = 5\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$.
- 14 Find the modulus and argument of $z = \frac{(1+i)^3}{1-i}$, using De Moivre's theorem for the numerator and polar division.
 $1 + i$ has modulus $\sqrt{2}$ and argument $\frac{\pi}{4}$, so $(1+i)^3$ has modulus $2\sqrt{2}$ and argument $\frac{3\pi}{4}$. $1 - i$ has modulus $\sqrt{2}$ and argument $-\frac{\pi}{4}$. Dividing: modulus $= \frac{2\sqrt{2}}{\sqrt{2}} = 2$; argument $= \frac{3\pi}{4} - \left(-\frac{\pi}{4}\right) = \pi$. (This means $z = -2$, purely real.)

De Moivre's theorem

- 15 Use De Moivre's theorem to find $\left[3\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)\right]^4$, giving your answer in Cartesian form.
 $3^4\left(\cos \frac{4\pi}{6} + i \sin \frac{4\pi}{6}\right) = 81\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right) = 81\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -\frac{81}{2} + \frac{81\sqrt{3}}{2}i$.
- 16 Use De Moivre's theorem to find $(1-i)^5$, giving your answer in Cartesian form.
 $|1-i| = \sqrt{2}$, argument $-\frac{\pi}{4}$. $(1-i)^5 = (\sqrt{2})^5\left(\cos\left(-\frac{5\pi}{4}\right) + i \sin\left(-\frac{5\pi}{4}\right)\right)$. Since $-\frac{5\pi}{4}$ is equivalent to $\frac{3\pi}{4}$: $= 4\sqrt{2}\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right) = -4 + 4i$.

Finding roots using De Moivre's theorem

- 17** Find the three cube roots of $z = 27\left(\cos \frac{\pi}{2} + i\sin \frac{\pi}{2}\right)$, giving arguments in $[0, 2\pi)$.
 Cube roots have modulus $27^{1/3} = 3$ and arguments $\frac{\pi/2 + 2\pi k}{3}$ for $k = 0, 1, 2$: $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$. Roots:
 $3\left(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6}\right), 3\left(\cos \frac{5\pi}{6} + i\sin \frac{5\pi}{6}\right), 3\left(\cos \frac{3\pi}{2} + i\sin \frac{3\pi}{2}\right)$.
- 18** Find all four fourth roots of -16 (i.e. solve $z^4 = -16$), giving arguments in $[0, 2\pi)$.
 Write $-16 = 16(\cos \pi + i\sin \pi)$. Fourth roots have modulus $16^{1/4} = 2$ and arguments $\frac{\pi + 2\pi k}{4}$ for $k = 0, 1, 2, 3$: $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$. Roots: $2\left(\cos \frac{\pi}{4} + i\sin \frac{\pi}{4}\right), 2\left(\cos \frac{3\pi}{4} + i\sin \frac{3\pi}{4}\right), 2\left(\cos \frac{5\pi}{4} + i\sin \frac{5\pi}{4}\right), 2\left(\cos \frac{7\pi}{4} + i\sin \frac{7\pi}{4}\right)$.

A well-designed word problem: AC circuit impedance

- 19** This question has three parts. An AC circuit has impedance $Z = R + iX_L$, where $R = 8 \Omega$ (resistance) and $X_L = 6 \Omega$ (inductive reactance). (a) Write Z in Cartesian form, find $|Z|$, and find $\arg Z$ to the nearest degree. (b) The current is $I = \frac{V}{Z}$, where $V = 100$ volts (real). Find I in the form $a + bi$, by rationalizing the denominator. (c) Find $|I|$, to two decimal places.
- (a) $Z = 8 + 6i$. $|Z| = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10 \Omega$. $\arg Z = \arctan\left(\frac{6}{8}\right) \approx 37^\circ$. (b)
 $I = \frac{100}{8 + 6i} = \frac{100(8 - 6i)}{(8 + 6i)(8 - 6i)} = \frac{100(8 - 6i)}{100} = 8 - 6i$ amps. (c)
 $|I| = \sqrt{8^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10.00$ amps, consistent with $|I| = \frac{|V|}{|Z|} = \frac{100}{10} = 10$.